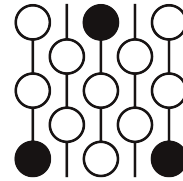


Review of Submission 6C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The solution is entirely correct, and the strategy is sensible but could perhaps be better signposted – reading this having already thought about the problem, it’s clear enough why something like Lemma 3 will be a key estimate but on first pass it’s a bit obscure. On a line-by-line level, all of the display equations seem to be true as written (not a bar that human-written math always clears!), but I didn’t feel that the balance of what to spell out and what to leave implicit was quite as well-tuned as in most publishable text.

Novelty. In outline, this solution follows the same approach as that of the problem authors: first analyze the trees that result from deleting the bad vertex, and then assemble everything together. The exact statements made about these subtrees do differ, but seemingly only superficially.

While the solution does find all the needed insights to solve the problem, to really call anything here novel I’d want to see a bit more – some amount of organizing these ideas into new concepts that would perhaps help with generalizations. After all, this problem is one case of a much bigger question, and the value in a solution is as much in ideas that could help with a full answer as it is in firmly nailing down this case. Such insights could perhaps be extracted from this answer, especially from the proof of Lemma 3, but this work has not been done by the writer. This makes it difficult to tell if there’s anything new here that isn’t also in e.g. the human-written solution to the problem.

The insight in this proof that seems most distinct from ideas in the human solution is the step in the proof of Lemma 3 in which D is bounded by the distance between τ and an integer – the human solution replaces this with a combinatorial argument for a stronger statement. Perhaps there is more that could be done with this idea, but distilling it properly from what has been written is not straightforward.

2 Recommendation

On a pure technical level, this has to be rated **essentially flawless**. All claims made are true, and none cannot be justified with any more than a bit of looking back in the text. I won’t say the exposition is perfect, but it certainly isn’t the most frustrating text I’ve ever read.

3 Submission with inline comments

Remark 1. Throughout, the phrase “a single vertex” with $w(v) < d_T(v)$ is interpreted as saying that v is the unique such vertex.

Lemma 1 (Norm-one elements are irreducible). *Let L be a positive definite integral lattice. If $x \in L$ satisfies $x^2 = 1$, then x is irreducible.*

Proof. Suppose $x = a + b$, where $a, b \in L$ are nonzero and $a \cdot b \geq 0$. Since L is positive definite and integral, $a^2, b^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = x^2 = a^2 + b^2 + 2a \cdot b \geq 1 + 1 = 2,$$

a contradiction. □

Let C be a rooted weighted tree with root ρ . Orient every edge away from ρ , and write $\text{ch}_C(x)$ for the number of children of a vertex $x \in C$. We call C admissible if its form is positive definite, every vertex weight is at least 2, and

$$w(x) \geq \text{ch}_C(x) + 1 \quad \text{for every } x \in C.$$

For $x = \sum n_y y \in L(C)$, write $x_y = n_y$ for the coefficient of the vertex y .

For a positive definite rooted tree C , define its capacity by

$$\gamma(C) = \left(Q_C^{-1}\right)_{\rho\rho},$$

where Q_C is the Gram matrix of C in the vertex basis.

Lemma 2 (Capacities of admissible rooted trees). *Let C be an admissible rooted tree with root ρ . Then*

$$0 < \gamma(C) < 1.$$

Moreover, if the root has children $\rho_1, \dots, \rho_t$, and C_i denotes the rooted subtree with root ρ_i , then

$$\gamma(C) = \frac{1}{w(\rho) - \sum_{i=1}^t \gamma(C_i)}.$$

Proof. Order the vertices so that ρ comes first and the remaining vertices are grouped by the rooted subtrees $C_1, \dots, C_t$. Then

$$Q_C = \begin{pmatrix} w(\rho) & c^T \\ c & Q_0 \end{pmatrix},$$

where $Q_0 = \bigoplus_i Q_{C_i}$, and c has entry -1 in the coordinate corresponding to each child root ρ_i and 0 elsewhere. By the Schur complement formula,

$$\left(Q_C^{-1}\right)_{\rho\rho} = \frac{1}{w(\rho) - c^T Q_0^{-1} c} = \frac{1}{w(\rho) - \sum_i \gamma(C_i)}.$$

It remains to show that $\gamma(C) < 1$. This follows by induction on the number of vertices. If C has one vertex, then $w(\rho) \geq 2$, so $\gamma(C) = 1/w(\rho) < 1$. Otherwise, by induction, $\gamma(C_i) < 1$ for every child subtree. If the root has t children, then admissibility gives $w(\rho) \geq t + 1$. Hence

$$w(\rho) - \sum_i \gamma(C_i) > w(\rho) - t \geq 1,$$

so $\gamma(C) < 1$. □

Reviewer 1

Could say why $\gamma(C) > 0$, but this is not hard.

Lemma 3 (A rooted estimate). *Let C be an admissible rooted tree with root ρ and capacity $\gamma = \gamma(C)$. Then, for every $x \in L(C)$ and every $k \in \mathbb{Z}$,*

$$x^2 - (2k + 1)x_\rho + \gamma k(k + 1) \geq 0.$$

Proof. We prove the claim by induction on $|C|$.

If C has one vertex, say $x = m\rho$, then $\gamma = 1/w(\rho)$. Let $W = w(\rho)$. Multiplying the desired inequality by W , we get

$$W \left(Wm^2 - (2k + 1)m + \frac{k(k + 1)}{W} \right) = (Wm - k)(Wm - k - 1).$$

Since $Wm - k \in \mathbb{Z}$, the product of the two consecutive integers $Wm - k$ and $Wm - k - 1$ is nonnegative. This proves the base case.

Now assume $|C| > 1$. Let the root be ρ , let its children be $\rho_1, \dots, \rho_t$, and let C_i be the rooted subtree with root ρ_i . Write

$$x = a\rho + \sum_i x_i, \quad x_i \in L(C_i),$$

and let $s_i = (x_i)_{\rho_i}$. Then

$$x^2 = w(\rho)a^2 - 2a \sum_i s_i + \sum_i x_i^2.$$

Let $\gamma_i = \gamma(C_i)$. By the induction hypothesis applied to C_i with $k = a$, we have

$$x_i^2 - (2a + 1)s_i + \gamma_i a(a + 1) \geq 0,$$

so

$$x_i^2 - 2as_i \geq s_i - \gamma_i a(a + 1).$$

Similarly, applying the induction hypothesis with $k = a - 1$, we obtain

$$x_i^2 - (2a - 1)s_i + \gamma_i a(a - 1) \geq 0,$$

so

$$x_i^2 - 2as_i \geq -s_i - \gamma_i a(a - 1).$$

Taking the maximum of these two lower bounds gives

$$x_i^2 - 2as_i \geq -\gamma_i a^2 + |s_i - \gamma_i a|.$$

Therefore, setting

$$D = \sum_i |s_i - \gamma_i a|,$$

we get

$$x^2 - (2k+1)a + \gamma k(k+1) \geq \left(w(\rho) - \sum_i \gamma_i \right) a^2 - (2k+1)a + \gamma k(k+1) + D.$$

By the capacity formula,

$$\gamma^{-1} = w(\rho) - \sum_i \gamma_i.$$

Thus, if $\tau = a/\gamma$, the preceding lower bound becomes

$$\gamma(\tau - k)(\tau - k - 1) + D.$$

If $\tau \notin (k, k+1)$, then $(\tau - k)(\tau - k - 1) \geq 0$, and the result follows. Suppose instead that $k < \tau < k+1$. Put

$$\alpha = \tau - k, \quad \beta = k+1 - \tau.$$

Then $0 < \alpha, \beta < 1$, $\alpha + \beta = 1$, and

$$\gamma(\tau - k)(\tau - k - 1) = -\gamma\alpha\beta.$$

Now

$$\tau = \frac{a}{\gamma} = \left(w(\rho) - \sum_i \gamma_i \right) a.$$

Since $w(\rho)a - \sum_i s_i \in \mathbb{Z}$, we have

$$D = \sum_i |s_i - \gamma_i a| \geq \text{dist}(\tau, \mathbb{Z}) = \min\{\alpha, \beta\}.$$

Reviewer 1

This seems to be a key step; I'd perhaps like it spelled out a little more but it is true.

Because $0 < \gamma < 1$ and $\alpha\beta \leq \min\{\alpha, \beta\}$, it follows that

$$D \geq \gamma\alpha\beta.$$

Hence

$$\gamma(\tau - k)(\tau - k - 1) + D \geq 0.$$

This completes the induction. □

Corollary 1. *Let C be an admissible rooted tree with root ρ . If $0 \neq x \in L(C)$, then*

$$x^2 - x_\rho > 0.$$

Proof. Let $\gamma = \gamma(C)$. If $x_\rho \leq 0$, then $x^2 - x_\rho > 0$ because C is positive definite and $x \neq 0$.

Assume $x_\rho > 0$. Let $\rho^\# \in L(C) \otimes \mathbb{Q}$ be the vector satisfying

$$\rho^\# \cdot y = y_\rho \quad \text{for all } y \in L(C).$$

Equivalently, the coefficient vector of $\rho^\#$ is $Q_C^{-1}e_\rho$. Thus

$$(\rho^\#)^2 = \gamma.$$

By Cauchy's inequality in the positive definite real vector space $L(C) \otimes \mathbb{R}$,

$$x_\rho^2 = (x \cdot \rho^\#)^2 \leq x^2(\rho^\#)^2 = \gamma x^2.$$

Since $\gamma < 1$, we get

$$x^2 \geq \frac{x_\rho^2}{\gamma} > x_\rho^2 \geq x_\rho.$$

Therefore $x^2 - x_\rho > 0$. □

Theorem 1. *Let T be a finite weighted tree whose associated form is positive definite. Suppose that there is a unique vertex v such that*

$$w(v) < d_T(v).$$

Then T contains a vertex that is irreducible in $L(T)$.

Proof. Since the form is positive definite, every vertex has positive square. Thus every vertex weight is a positive integer.

If some vertex u has $w(u) = 1$, then $u^2 = 1$, and u is irreducible by the first lemma. Hence we may assume from now on that every vertex weight is at least 2.

Let v be the unique vertex with $w(v) < d_T(v)$. We will prove that v itself is irreducible.

Let $C_1, \dots, C_m$ be the connected components of $T \setminus \{v\}$. For each i , let ρ_i be the unique vertex of C_i adjacent to v , and root C_i at ρ_i . Because v is the unique bad vertex, every vertex $x \in C_i$ satisfies

$$w(x) \geq d_T(x).$$

With the orientation away from ρ_i , each $x \in C_i$ has exactly one edge pointing toward v or toward its parent, so

$$d_T(x) = \text{ch}_{C_i}(x) + 1.$$

Therefore

$$w(x) \geq \text{ch}_{C_i}(x) + 1.$$

Since all weights are at least 2, each C_i is admissible. Let

$$\gamma_i = \gamma(C_i).$$

By the Schur complement formula applied to the decomposition of T into v and the components C_i , positive definiteness of T implies

$$w(v) - \sum_i \gamma_i > 0.$$

Write $W = w(v)$.

Suppose, toward a contradiction, that v is reducible. Then there exist nonzero $a, b \in L(T)$ such that

$$v = a + b, \quad a \cdot b \geq 0.$$

Put $z = -b$. Then $a = v + z$, and

$$0 \leq a \cdot b = (v + z) \cdot (-z) = -(z^2 + v \cdot z).$$

Hence

$$z^2 + v \cdot z \leq 0.$$

Also $z \neq 0$ and $z \neq -v$, because $a, b \neq 0$.

Let q be the coefficient of v in z . If $q \leq -1$, replace z by

$$z' = -v - z.$$

This replacement corresponds to interchanging a and b , and

$$z' \cdot (z' + v) = z \cdot (z + v).$$

Moreover, the coefficient of v in z' is $-1 - q \geq 0$. Thus we may assume that the coefficient of v in z is a nonnegative integer p .

Write

$$z = pv + \sum_i y_i, \quad y_i \in L(C_i).$$

Let $s_i = (y_i)_{\rho_i}$. Since $v \cdot y_i = -s_i$, we compute

$$\begin{aligned} z^2 + v \cdot z &= \left(Wp^2 - 2p \sum_i s_i + \sum_i y_i^2 \right) + \left(Wp - \sum_i s_i \right) \\ &= Wp(p+1) + \sum_i (y_i^2 - (2p+1)s_i). \end{aligned}$$

First suppose $p \geq 1$. Applying the rooted estimate to each C_i with $k = p$, we get

$$y_i^2 - (2p+1)s_i \geq -\gamma_i p(p+1).$$

Therefore

$$z^2 + v \cdot z \geq \left(W - \sum_i \gamma_i \right) p(p+1) > 0,$$

contradicting $z^2 + v \cdot z \leq 0$.

It remains to consider $p = 0$. Then

$$z^2 + v \cdot z = \sum_i (y_i^2 - s_i).$$

By the corollary, each term $y_i^2 - s_i$ is nonnegative, and it is strictly positive whenever $y_i \neq 0$. Since $z \neq 0$ and $p = 0$, at least one y_i is nonzero. Hence

$$z^2 + v \cdot z > 0,$$

again contradicting $z^2 + v \cdot z \leq 0$.

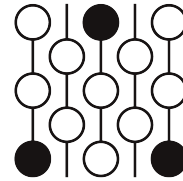
Thus v is irreducible. Consequently, T contains an irreducible vertex. □

Review of Submission 6C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The proof is correct and reasonably well explained, although the writing has various issues. The definitions are repetitive, there's one place where it summarizes the proof of a lemma rather than using the lemma, and the notation is not always well chosen. The proof also explains some easy steps (such as the inequality comparing w with $\text{ch}_C + 1$ in the proof of Theorem 1) in detail, while going through more difficult things quickly.

I expect the proof could be made more conceptual with some reformulation, maybe in terms of dual lattices, but it is correct and explained enough to be checkable.

Novelty. I don't know to what extent this argument takes ideas from the literature. A few things make me wonder, such as the term "capacity," which hints at connections beyond the problem statement. I don't know whether this is just because LLMs like to name things and choose a vague but evocative name when they can't come up with a standard name, or whether it's an indication of some connection.

2 Recommendation

Mathematically, it is essentially flawless and does not require any factual corrections. Conceptually, I think it could likely be clarified, but a first proof is often clumsy. Stylistically, I would suggest a bunch of changes before publication.

With a human author, I would view this level of editing as minor revisions not requiring further review, but I'd be a little more skeptical of AI's ability to carry out the revisions well. In any case, these revisions would not be mathematical corrections.

3 Submission with inline comments

Remark 1. *Throughout, the phrase "a single vertex" with $w(v) < d_T(v)$ is interpreted as saying that v is the unique such vertex.*

The notation is mutating slightly: what was d in the problem statement is now d_T .

Lemma 1 (Norm-one elements are irreducible). *Let L be a positive definite integral lattice. If $x \in L$ satisfies $x^2 = 1$, then x is irreducible.*

Proof. Suppose $x = a + b$, where $a, b \in L$ are nonzero and $a \cdot b \geq 0$. Since L is positive definite and integral, $a^2, b^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = x^2 = a^2 + b^2 + 2a \cdot b \geq 1 + 1 = 2,$$

a contradiction. □

Let C be a rooted weighted tree with root ρ . Orient every edge away from ρ , and write $\text{ch}_C(x)$ for the number of children of a vertex $x \in C$. We call C admissible if its form is positive definite, every vertex weight is at least 2, and

$$w(x) \geq \text{ch}_C(x) + 1 \quad \text{for every } x \in C.$$

For $x = \sum n_y y \in L(C)$, write $x_y = n_y$ for the coefficient of the vertex y .

For a positive definite rooted tree C , define its capacity by

$$\gamma(C) = (Q_C^{-1})_{\rho\rho},$$

where Q_C is the Gram matrix of C in the vertex basis.

Is “capacity” a standard term here? If so, this should be said, preferably with a reference. For example, does it come from electrical networks and random walks?

Lemma 2 (Capacities of admissible rooted trees). *Let C be an admissible rooted tree with root ρ . Then*

$$0 < \gamma(C) < 1.$$

Moreover, if the root has children $\rho_1, \dots, \rho_t$, and C_i denotes the rooted subtree with root ρ_i , then

$$\gamma(C) = \frac{1}{w(\rho) - \sum_{i=1}^t \gamma(C_i)}.$$

Proof. Order the vertices so that ρ comes first and the remaining vertices are grouped by the rooted subtrees $C_1, \dots, C_t$. Then

$$Q_C = \begin{pmatrix} w(\rho) & c^T \\ c & Q_0 \end{pmatrix},$$

where $Q_0 = \bigoplus_i Q_{C_i}$, and c has entry -1 in the coordinate corresponding to each child root ρ_i and 0 elsewhere. By the Schur complement formula,

$$(Q_C^{-1})_{\rho\rho} = \frac{1}{w(\rho) - c^T Q_0^{-1} c} = \frac{1}{w(\rho) - \sum_i \gamma(C_i)}.$$

It remains to show that $\gamma(C) < 1$. This follows by induction on the number of vertices. If C has one vertex, then $w(\rho) \geq 2$, so $\gamma(C) = 1/w(\rho) < 1$. Otherwise, by induction, $\gamma(C_i) < 1$ for every child subtree. If the root has t children, then admissibility gives $w(\rho) \geq t + 1$. Hence

$$w(\rho) - \sum_i \gamma(C_i) > w(\rho) - t \geq 1,$$

so $\gamma(C) < 1$. □

Lemma 3 (A rooted estimate). *Let C be an admissible rooted tree with root ρ and capacity $\gamma = \gamma(C)$. Then, for every $x \in L(C)$ and every $k \in \mathbb{Z}$,*

$$x^2 - (2k + 1)x_\rho + \gamma k(k + 1) \geq 0.$$

Proof. We prove the claim by induction on $|C|$.

If C has one vertex, say $x = m\rho$, then $\gamma = 1/w(\rho)$. Let $W = w(\rho)$. Multiplying the desired inequality by W , we get

$$W \left(Wm^2 - (2k + 1)m + \frac{k(k + 1)}{W} \right) = (Wm - k)(Wm - k - 1).$$

Since $Wm - k \in \mathbb{Z}$, the product of the two consecutive integers $Wm - k$ and $Wm - k - 1$ is nonnegative. This proves the base case.

Now assume $|C| > 1$. Let the root be ρ , let its children be $\rho_1, \dots, \rho_t$, and let C_i be the rooted subtree with root ρ_i . Write

$$x = a\rho + \sum_i x_i, \quad x_i \in L(C_i),$$

and let $s_i = (x_i)_{\rho_i}$. Then

$$x^2 = w(\rho)a^2 - 2a \sum_i s_i + \sum_i x_i^2.$$

Let $\gamma_i = \gamma(C_i)$. By the induction hypothesis applied to C_i with $k = a$, we have

$$x_i^2 - (2a + 1)s_i + \gamma_i a(a + 1) \geq 0,$$

so

$$x_i^2 - 2as_i \geq s_i - \gamma_i a(a + 1).$$

Similarly, applying the induction hypothesis with $k = a - 1$, we obtain

$$x_i^2 - (2a - 1)s_i + \gamma_i a(a - 1) \geq 0,$$

so

$$x_i^2 - 2as_i \geq -s_i - \gamma_i a(a - 1).$$

Taking the maximum of these two lower bounds gives

$$x_i^2 - 2as_i \geq -\gamma_i a^2 + |s_i - \gamma_i a|.$$

Therefore, setting

$$D = \sum_i |s_i - \gamma_i a|,$$

we get

$$x^2 - (2k+1)a + \gamma k(k+1) \geq \left(w(\rho) - \sum_i \gamma_i\right) a^2 - (2k+1)a + \gamma k(k+1) + D.$$

By the capacity formula,

$$\gamma^{-1} = w(\rho) - \sum_i \gamma_i.$$

Thus, if $\tau = a/\gamma$, the preceding lower bound becomes

$$\gamma(\tau - k)(\tau - k - 1) + D.$$

If $\tau \notin (k, k+1)$, then $(\tau - k)(\tau - k - 1) \geq 0$, and the result follows. Suppose instead that $k < \tau < k+1$. Put

$$\alpha = \tau - k, \quad \beta = k + 1 - \tau.$$

Then $0 < \alpha, \beta < 1$, $\alpha + \beta = 1$, and

$$\gamma(\tau - k)(\tau - k - 1) = -\gamma\alpha\beta.$$

Now

$$\tau = \frac{a}{\gamma} = \left(w(\rho) - \sum_i \gamma_i\right) a.$$

Since $w(\rho)a - \sum_i s_i \in \mathbb{Z}$, we have

$$D = \sum_i |s_i - \gamma_i a| \geq \text{dist}(\tau, \mathbb{Z}) = \min\{\alpha, \beta\}.$$

Because $0 < \gamma < 1$ and $\alpha\beta \leq \min\{\alpha, \beta\}$, it follows that

$$D \geq \gamma\alpha\beta.$$

Hence

$$\gamma(\tau - k)(\tau - k - 1) + D \geq 0.$$

This completes the induction. □

Corollary 1. *Let C be an admissible rooted tree with root ρ . If $0 \neq x \in L(C)$, then*

$$x^2 - x_\rho > 0.$$

Proof. Let $\gamma = \gamma(C)$. If $x_\rho \leq 0$, then $x^2 - x_\rho > 0$ because C is positive definite and $x \neq 0$. Assume $x_\rho > 0$. Let $\rho^\# \in L(C) \otimes \mathbb{Q}$ be the vector satisfying

$$\rho^\# \cdot y = y_\rho \quad \text{for all } y \in L(C).$$

Equivalently, the coefficient vector of $\rho^\#$ is $Q_C^{-1} e_\rho$. Thus

$$(\rho^\#)^2 = \gamma.$$

By Cauchy's inequality in the positive definite real vector space $L(C) \otimes \mathbb{R}$,

$$x_\rho^2 = (x \cdot \rho^\#)^2 \leq x^2 (\rho^\#)^2 = \gamma x^2.$$

Since $\gamma < 1$, we get

$$x^2 \geq \frac{x_\rho^2}{\gamma} > x_\rho^2 \geq x_\rho.$$

Therefore $x^2 - x_\rho > 0$. □

Theorem 1. *Let T be a finite weighted tree whose associated form is positive definite. Suppose that there is a unique vertex v such that*

$$w(v) < d_T(v).$$

Then T contains a vertex that is irreducible in $L(T)$.

Proof. Since the form is positive definite, every vertex has positive square. Thus every vertex weight is a positive integer.

If some vertex u has $w(u) = 1$, then $u^2 = 1$, and u is irreducible by the first lemma. Hence we may assume from now on that every vertex weight is at least 2.

Let v be the unique vertex with $w(v) < d_T(v)$. We will prove that v itself is irreducible.

Let $C_1, \dots, C_m$ be the connected components of $T \setminus \{v\}$. For each i , let ρ_i be the unique vertex of C_i adjacent to v , and root C_i at ρ_i . Because v is the unique bad vertex, every vertex $x \in C_i$ satisfies

$$w(x) \geq d_T(x).$$

With the orientation away from ρ_i , each $x \in C_i$ has exactly one edge pointing toward v or toward its parent, so

$$d_T(x) = \text{ch}_{C_i}(x) + 1.$$

Therefore

$$w(x) \geq \text{ch}_{C_i}(x) + 1.$$

Since all weights are at least 2, each C_i is admissible. Let

$$\gamma_i = \gamma(C_i).$$

By the Schur complement formula applied to the decomposition of T into v and the components C_i , positive definiteness of T implies

$$w(v) - \sum_i \gamma_i > 0.$$

It's better to cite Lemma 2 than just briefly recapitulate the proof.

Write $W = w(v)$.

Suppose, toward a contradiction, that v is reducible. Then there exist nonzero $a, b \in L(T)$ such that

$$v = a + b, \quad a \cdot b \geq 0.$$

Put $z = -b$. Then $a = v + z$, and

$$0 \leq a \cdot b = (v + z) \cdot (-z) = -(z^2 + v \cdot z).$$

Hence

$$z^2 + v \cdot z \leq 0.$$

Also $z \neq 0$ and $z \neq -v$, because $a, b \neq 0$.

Let q be the coefficient of v in z . If $q \leq -1$, replace z by

$$z' = -v - z.$$

This replacement corresponds to interchanging a and b , and

$$z' \cdot (z' + v) = z \cdot (z + v).$$

Moreover, the coefficient of v in z' is $-1 - q \geq 0$. Thus we may assume that the coefficient of v in z is a nonnegative integer p .

You were just calling it q , and now switch to p for no apparent reason.

Write

$$z = pv + \sum_i y_i, \quad y_i \in L(C_i).$$

Let $s_i = (y_i)_{\rho_i}$. Since $v \cdot y_i = -s_i$, we compute

$$\begin{aligned} z^2 + v \cdot z &= \left(Wp^2 - 2p \sum_i s_i + \sum_i y_i^2 \right) + \left(Wp - \sum_i s_i \right) \\ &= Wp(p+1) + \sum_i \left(y_i^2 - (2p+1)s_i \right). \end{aligned}$$

First suppose $p \geq 1$. Applying the rooted estimate to each C_i with $k = p$, we get

$$y_i^2 - (2p+1)s_i \geq -\gamma_i p(p+1).$$

Therefore

$$z^2 + v \cdot z \geq \left(W - \sum_i \gamma_i \right) p(p+1) > 0,$$

contradicting $z^2 + v \cdot z \leq 0$.

It remains to consider $p = 0$. Then

$$z^2 + v \cdot z = \sum_i (y_i^2 - s_i).$$

By the corollary, each term $y_i^2 - s_i$ is nonnegative, and it is strictly positive whenever $y_i \neq 0$. Since $z \neq 0$ and $p = 0$, at least one y_i is nonzero. Hence

$$z^2 + v \cdot z > 0,$$

again contradicting $z^2 + v \cdot z \leq 0$.

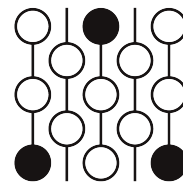
Thus v is irreducible. Consequently, T contains an irreducible vertex. □

Review of Submission 6C

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

This gives a complete solution to the problem. The approach contains essentially all the same ideas as the known human proof. Although these ideas are more obscured in the presentation here, in the sense that the proof makes a direct beeline to the solution using elementary algebraic manipulation and does not provide any motivation or interpretation of the steps. Although argument appears to be correct in all aspects, I suspect that certain parts of the argument could be simplified.

Exposition In terms of the writing, I was not convinced by some of the notational and stylistic choices. Although it is readable, I would certainly have a lot to say if this were written by a grad student of mine.

Correctness. The solution is correct.

Novelty. This solution roughly relies on the same ideas as the known human solution, but with the conceptual underpinnings stripped out. This results in an argument is extremely specialized to the case at hand and does not appear to contain any new insights that could be directly generalized to related problems. The key inequality, Lemma 3, is a delicate induction argument. I suspect it unlikely (but not impossible) for a human to prove the the same inequality in this manner.

2 Recommendation

Based on the provided rubric, I would rate the solution to be essentially flawless. Although I do not think it is well written, it contains a complete solution in which all the details appear correct.

3 Submission with inline comments

Remark 1. *Throughout, the phrase “a single vertex” with $w(v) < d_T(v)$ is interpreted as saying that v is the unique such vertex.*

Lemma 1 (Norm-one elements are irreducible). *Let L be a positive definite integral lattice. If $x \in L$ satisfies $x^2 = 1$, then x is irreducible.*

Proof. Suppose $x = a + b$, where $a, b \in L$ are nonzero and $a \cdot b \geq 0$. Since L is positive definite and integral, $a^2, b^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = x^2 = a^2 + b^2 + 2a \cdot b \geq 1 + 1 = 2,$$

a contradiction. □

Let C be a rooted weighted tree with root ρ . Orient every edge away from ρ , and write $\text{ch}_C(x)$ for the number of children of a vertex $x \in C$. We call C admissible if its form is positive definite, every vertex weight is at least 2, and

$$w(x) \geq \text{ch}_C(x) + 1 \quad \text{for every } x \in C.$$

For $x = \sum n_y y \in L(C)$, write $x_y = n_y$ for the coefficient of the vertex y .

For a positive definite rooted tree C , define its capacity by

$$\gamma(C) = \left(Q_C^{-1} \right)_{\rho\rho},$$

where Q_C is the Gram matrix of C in the vertex basis.

Lemma 2 (Capacities of admissible rooted trees). *Let C be an admissible rooted tree with root ρ . Then*

$$0 < \gamma(C) < 1.$$

Moreover, if the root has children $\rho_1, \dots, \rho_t$, and C_i denotes the rooted subtree with root ρ_i , then

$$\gamma(C) = \frac{1}{w(\rho) - \sum_{i=1}^t \gamma(C_i)}.$$

Proof. Order the vertices so that ρ comes first and the remaining vertices are grouped by the rooted subtrees $C_1, \dots, C_t$. Then

$$Q_C = \begin{pmatrix} w(\rho) & c^T \\ c & Q_0 \end{pmatrix},$$

where $Q_0 = \bigoplus_i Q_{C_i}$, and c has entry -1 in the coordinate corresponding to each child root ρ_i and 0 elsewhere. By the Schur complement formula,

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I'm not entirely what is meant by Schur complement formula here. I am not familiar with that nomenclature. The identity is true though.

$$\left(Q_C^{-1} \right)_{\rho\rho} = \frac{1}{w(\rho) - c^T Q_0^{-1} c} = \frac{1}{w(\rho) - \sum_i \gamma(C_i)}.$$

It remains to show that $\gamma(C) < 1$. This follows by induction on the number of vertices. If C has one vertex, then $w(\rho) \geq 2$, so $\gamma(C) = 1/w(\rho) < 1$. Otherwise, by induction, $\gamma(C_i) < 1$ for every child subtree. If the root has t children, then admissibility gives $w(\rho) \geq t + 1$. Hence

$$w(\rho) - \sum_i \gamma(C_i) > w(\rho) - t \geq 1,$$

so $\gamma(C) < 1$. □

Lemma 3 (A rooted estimate). *Let C be an admissible rooted tree with root ρ and capacity $\gamma = \gamma(C)$. Then, for every $x \in L(C)$ and every $k \in \mathbb{Z}$,*

$$x^2 - (2k + 1)x_\rho + \gamma k(k + 1) \geq 0.$$

Proof. We prove the claim by induction on $|C|$.

If C has one vertex, say $x = m\rho$, then $\gamma = 1/w(\rho)$. Let $W = w(\rho)$. Multiplying the desired inequality by W , we get

$$W \left(Wm^2 - (2k + 1)m + \frac{k(k + 1)}{W} \right) = (Wm - k)(Wm - k - 1).$$

Since $Wm - k \in \mathbb{Z}$, the product of the two consecutive integers $Wm - k$ and $Wm - k - 1$ is nonnegative. This proves the base case.

Now assume $|C| > 1$. Let the root be ρ , let its children be $\rho_1, \dots, \rho_t$, and let C_i be the rooted subtree with root ρ_i . Write

$$x = a\rho + \sum_i x_i, \quad x_i \in L(C_i),$$

and let $s_i = (x_i)_{\rho_i}$. Then

$$x^2 = w(\rho)a^2 - 2a \sum_i s_i + \sum_i x_i^2.$$

Let $\gamma_i = \gamma(C_i)$. By the induction hypothesis applied to C_i with $k = a$, we have

$$x_i^2 - (2a + 1)s_i + \gamma_i a(a + 1) \geq 0,$$

so

$$x_i^2 - 2as_i \geq s_i - \gamma_i a(a + 1).$$

Similarly, applying the induction hypothesis with $k = a - 1$, we obtain

$$x_i^2 - (2a - 1)s_i + \gamma_i a(a - 1) \geq 0,$$

so

$$x_i^2 - 2as_i \geq -s_i - \gamma_i a(a - 1).$$

Taking the maximum of these two lower bounds gives

$$x_i^2 - 2as_i \geq -\gamma_i a^2 + |s_i - \gamma_i a|.$$

Therefore, setting

$$D = \sum_i |s_i - \gamma_i a|,$$

we get

$$x^2 - (2k + 1)a + \gamma k(k + 1) \geq \left(w(\rho) - \sum_i \gamma_i \right) a^2 - (2k + 1)a + \gamma k(k + 1) + D.$$

By the capacity formula,

$$\gamma^{-1} = w(\rho) - \sum_i \gamma_i.$$

Thus, if $\tau = a/\gamma$, the preceding lower bound becomes

$$\gamma(\tau - k)(\tau - k - 1) + D.$$

If $\tau \notin (k, k + 1)$, then $(\tau - k)(\tau - k - 1) \geq 0$, and the result follows. Suppose instead that $k < \tau < k + 1$. Put

$$\alpha = \tau - k, \quad \beta = k + 1 - \tau.$$

Then $0 < \alpha, \beta < 1$, $\alpha + \beta = 1$, and

$$\gamma(\tau - k)(\tau - k - 1) = -\gamma\alpha\beta.$$

Now

$$\tau = \frac{a}{\gamma} = \left(w(\rho) - \sum_i \gamma_i \right) a.$$

Since $w(\rho)a - \sum_i s_i \in \mathbb{Z}$, we have

$$D = \sum_i |s_i - \gamma_i a| \geq \text{dist}(\tau, \mathbb{Z}) = \min\{\alpha, \beta\}.$$

Because $0 < \gamma < 1$ and $\alpha\beta \leq \min\{\alpha, \beta\}$, it follows that

$$D \geq \gamma\alpha\beta.$$

Hence

$$\gamma(\tau - k)(\tau - k - 1) + D \geq 0.$$

This completes the induction. □

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I am not sure why this is named as a corollary. It does not appear to actually depend on preceding results.

Corollary 1. *Let C be an admissible rooted tree with root ρ . If $0 \neq x \in L(C)$, then*

$$x^2 - x_\rho > 0.$$

Proof. Let $\gamma = \gamma(C)$. If $x_\rho \leq 0$, then $x^2 - x_\rho > 0$ because C is positive definite and $x \neq 0$. Assume $x_\rho > 0$. Let $\rho^\# \in L(C) \otimes \mathbb{Q}$ be the vector satisfying

$$\rho^\# \cdot y = y_\rho \quad \text{for all } y \in L(C).$$

Equivalently, the coefficient vector of $\rho^\#$ is $Q_C^{-1}e_\rho$. Thus

$$(\rho^\#)^2 = \gamma.$$

By Cauchy's inequality in the positive definite real vector space $L(C) \otimes \mathbb{R}$,

$$x_\rho^2 = (x \cdot \rho^\#)^2 \leq x^2(\rho^\#)^2 = \gamma x^2.$$

Since $\gamma < 1$, we get

$$x^2 \geq \frac{x_\rho^2}{\gamma} > x_\rho^2 \geq x_\rho.$$

Therefore $x^2 - x_\rho > 0$. □

Theorem 1. *Let T be a finite weighted tree whose associated form is positive definite. Suppose that there is a unique vertex v such that*

$$w(v) < d_T(v).$$

Then T contains a vertex that is irreducible in $L(T)$.

Proof. Since the form is positive definite, every vertex has positive square. Thus every vertex weight is a positive integer.

If some vertex u has $w(u) = 1$, then $u^2 = 1$, and u is irreducible by the first lemma. Hence we may assume from now on that every vertex weight is at least 2.

Let v be the unique vertex with $w(v) < d_T(v)$. We will prove that v itself is irreducible.

Let $C_1, \dots, C_m$ be the connected components of $T \setminus \{v\}$. For each i , let ρ_i be the unique vertex of C_i adjacent to v , and root C_i at ρ_i . Because v is the unique bad vertex, every vertex $x \in C_i$ satisfies

$$w(x) \geq d_T(x).$$

With the orientation away from ρ_i , each $x \in C_i$ has exactly one edge pointing toward v or toward its parent, so

$$d_T(x) = \text{ch}_{C_i}(x) + 1.$$

Therefore

$$w(x) \geq \text{ch}_{C_i}(x) + 1.$$

Since all weights are at least 2, each C_i is admissible. Let

$$\gamma_i = \gamma(C_i).$$

By the Schur complement formula applied to the decomposition of T into v and the components C_i , positive definiteness of T implies

$$w(v) - \sum_i \gamma_i > 0.$$

Write $W = w(v)$.

Suppose, toward a contradiction, that v is reducible. Then there exist nonzero $a, b \in L(T)$ such that

$$v = a + b, \quad a \cdot b \geq 0.$$

Put $z = -b$. Then $a = v + z$, and

$$0 \leq a \cdot b = (v + z) \cdot (-z) = -(z^2 + v \cdot z).$$

Hence

$$z^2 + v \cdot z \leq 0.$$

Also $z \neq 0$ and $z \neq -v$, because $a, b \neq 0$.

Let q be the coefficient of v in z . If $q \leq -1$, replace z by

$$z' = -v - z.$$

This replacement corresponds to interchanging a and b , and

$$z' \cdot (z' + v) = z \cdot (z + v).$$

Moreover, the coefficient of v in z' is $-1 - q \geq 0$. Thus we may assume that the coefficient of v in z is a nonnegative integer p .

Write

$$z = pv + \sum_i y_i, \quad y_i \in L(C_i).$$

Let $s_i = (y_i)_{\rho_i}$. Since $v \cdot y_i = -s_i$, we compute

$$\begin{aligned} z^2 + v \cdot z &= \left(Wp^2 - 2p \sum_i s_i + \sum_i y_i^2 \right) + \left(Wp - \sum_i s_i \right) \\ &= Wp(p+1) + \sum_i (y_i^2 - (2p+1)s_i). \end{aligned}$$

First suppose $p \geq 1$. Applying the rooted estimate to each C_i with $k = p$, we get

$$y_i^2 - (2p+1)s_i \geq -\gamma_i p(p+1).$$

Therefore

$$z^2 + v \cdot z \geq \left(W - \sum_i \gamma_i \right) p(p+1) > 0,$$

contradicting $z^2 + v \cdot z \leq 0$.

It remains to consider $p = 0$. Then

$$z^2 + v \cdot z = \sum_i (y_i^2 - s_i).$$

By the corollary, each term $y_i^2 - s_i$ is nonnegative, and it is strictly positive whenever $y_i \neq 0$. Since $z \neq 0$ and $p = 0$, at least one y_i is nonzero. Hence

$$z^2 + v \cdot z > 0,$$

again contradicting $z^2 + v \cdot z \leq 0$.

Thus v is irreducible. Consequently, T contains an irreducible vertex. □